Campus Network Design using Cisco Packet Tracer

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1 Project Overview

This project demonstrates the design and simulation of a multi-subnet campus network using Cisco Packet Tracer. The objective is to create a structured and scalable network environment with proper segmentation, routing, and automatic IP allocation.

2 Objectives

- Design a multi-subnet campus network
- Implement VLAN segmentation for departments
- Configure inter-VLAN routing using router-on-a-stick
- Enable DHCP for automatic IP assignment
- Test and verify network connectivity

3 Network Design

The network consists of:

- 1 Router (Inter-VLAN Routing)
- 1 Switch (Layer 2)
- 3 PCs representing departments: Admin, Staff, Students

Each department is assigned a separate VLAN to ensure logical segmentation and improved network management.

4 VLAN Configuration

- VLAN 10: Admin Department
- VLAN 20: Staff Department
- VLAN 30: Students Department

Each VLAN is mapped to specific switch ports.

5 Inter-VLAN Routing

Router-on-a-stick configuration was implemented using sub-interfaces:

- VLAN 10 → 192.168.10.1/24
- VLAN 20 → 192.168.20.1/24
- VLAN 30 → 192.168.30.1/24

IEEE 802.1Q encapsulation was used for VLAN tagging.

6 DHCP Configuration

A DHCP server was configured on the router to automatically assign IP addresses for all VLANs, reducing manual configuration and improving efficiency.

7 Testing and Results

Network connectivity was tested using ping commands. All devices successfully communicated within and across VLANs, confirming proper configuration of VLANs, routing, and DHCP.

8 Conclusion

This project successfully demonstrates a functional campus network using Cisco Packet Tracer. It provides practical experience in VLAN configuration, inter-VLAN routing, and DHCP setup, which are essential skills for network engineering roles.

9 Tools Used

- Cisco Packet Tracer
- VLAN (IEEE 802.1Q)
- DHCP
- Router-on-a-stick
- TCP/IP Networking